

Technical Assignment: Mall Footfall Data Analysis

Project: Sensor-based footfall modelling and visitor behaviour analysis

Data period: 7 days (April 20-26, 2026)

Location: One shopping mall (Liverpool, UK)

Background & Context

AllUnite provides audience measurement solutions for out-of-home (OOH) advertising. The core idea is simple: sensors installed at different locations inside a mall detect Wi-Fi and Bluetooth signals from people's smartphones as they walk by. Each time a device is detected, a **session** is recorded — capturing which sensor detected it, when, and for how long.

However, **raw device detections ≠ real visitor counts**.

Smartphones behave unpredictably: some people have Wi-Fi turned off, some devices appear multiple times, and sensors can pick up background “noise” from nearby residents or parked cars. To convert raw sensor data into actual footfall numbers, we use **Manual Counting (MC)** — human observers who count real people passing a sensor for a fixed time window. This gives us a ground-truth reference that we use to calibrate the sensor data.

Your job is to work with both data sources and perform the analyses described below.

Files Provided

1. allunite_device_session.csv — Raw Sensor Sessions

~2.25 million rows. Each row represents one detection event: a sensor picked up a signal from one device at a specific moment.

Column	Type	Description
session_start	datetime	Timestamp when the device was first detected in this session
device_id	int64	Anonymised unique identifier for the detected device (phone). Not a real device ID — hashed for privacy
company_num	int64	Numeric ID of the client company that owns the mall. All rows will share the same value in this dataset
facility_num	int64	Numeric ID of the specific

		sensor/facility that detected this device
facility_master_num	int64	Numeric ID of the mall as a whole (groups all sensors together). All rows share one value here
box_mac	string	MAC address of the physical sensor box. Each facility has one box; this links sessions to hardware
session_duration	int64	How long (in seconds) the device was continuously detected by this sensor. 0 means it was seen only momentarily
signal_level	int64	RSSI signal strength in dBm (e.g. -87). Higher absolute value = weaker signal = device is farther away
vendor_id	int64	Device manufacturer identifier derived from MAC address prefix. 0 = unknown or randomised MAC
is_permanent_device	bool	True if this device appears very frequently and is likely a fixed/permanent device (e.g. a staff member's phone or a shop's till)
is_local	bool	True if the device has a locally administered MAC address, meaning its MAC address is randomised by the operating system (common on modern iOS and Android devices). These devices are harder to track consistently across sessions
is_trusted	bool	True if the device has a stable, non-randomised MAC address (more reliable for tracking)
is_excluded	bool	True if this device has been flagged for exclusion from traffic calculations (e.g. staff, known noise)
		True if this session

is_anomaly	bool	was algorithmically flagged as anomalous
is_fake	bool	True if this session is suspected to be a fake/spoofed signal

2. facility_information_-_Sheet1.csv — Sensor/Facility Metadata

10 rows. One row per sensor location installed in the mall. All sensors belong to the same mall (facility_master_num = 66328).

Column	Type	Description
facility_id	string (UUID)	Unique string identifier for the facility
company_id	string (UUID)	Unique string identifier for the company
facility_name	string	Internal name/code for the facility (e.g. GB-LV0-DD-08003)
facility_address	string	Physical street address of the sensor
latitude / longitude	float	GPS coordinates of the sensor placement
company_num	int64	Numeric company ID (matches session data)
facility_num	int64	Numeric facility ID (matches session data)
facility_master_num	int64	Numeric mall ID (matches session data)
asset_category	string	Category tag (e.g. Mall_2026)
box_macs	string (JSON list)	MAC address(es) of the sensor box(es) at this location (matches box_mac in sessions)
frame_count	int64	Number of advertising frames/screens at this location
facility_type	string	Type of asset (e.g. static)
is_active	int	1 = sensor is currently active

Use this file to understand the spatial layout of sensors, map facility_num to names and coordinates, and identify which box_mac belongs to which location.

3. Manual_Counting_-_Sheet1.csv — Ground Truth Counts

8 rows. Human observers stood near specific sensors and manually counted everyone who passed during a 60-minute window. This is the ground truth data used to calibrate sensor readings.

Column	Type	Description
id	string (UUID)	Unique ID for this counting session
name	string	Human-readable label for this count
facility_id	string (UUID)	Links to facility_information: which sensor location was counted
facility_num	int64	Numeric facility ID (matches session data)
facility_name	string	Name of the facility where counting took place
is_validated	bool	Whether this count was reviewed and approved
is_used	bool	Whether this count has already been used in a calibration run
started	datetime (UTC)	When the counting period started
duration	int64	Duration of the counting period in minutes (always 60 in this dataset)
comments	string	Notes from the counter, e.g. "Wi-fi - 5m." = the counter stood ~5 metres from the Wi-Fi sensor
category	string	What was counted: Pedestrians or Bicycles
count	int64	The number of people (or bicycles) counted during the window

Tasks

Task 1 — Traffic Model: Estimate Real Visitor Count from Sensor Data

Goal: Build a model that takes raw session data as input and outputs an estimated real footfall number, calibrated against the manual counts.

Context: The sensor detects many devices, but not all represent unique passing visitors. Some devices are seen repeatedly, some belong to staff, some are noise. The manual counts give you the “real”

answer for specific 1-hour windows at 2 sensor locations. Your task is to find the relationship between sensor signal and actual footfall.

Task 2 — Sensor Intersection: How Many Devices Were Seen by Multiple Sensors?

Goal: Understand how devices move between sensor locations, and what overlap exists between sensors.

Context: A single visitor walking through the mall will likely be detected by multiple sensors as they move. Identifying the same device across sensors tells us about movement paths and sensor coverage overlap.

Task 3 — Journey Mapping: Reconstruct Visitor Paths Through the Mall

Goal: For devices seen at multiple sensors, reconstruct the sequence of locations visited — i.e. the visitor's journey through the mall.

Context: We will provide you with a floor plan image of the mall showing where each sensor is located. Your task is to trace device paths across sensors ordered by time.

Task 4 — Anomaly Detection & Pattern Analysis

Goal: Identify unusual patterns, anomalies, and interesting behavioural trends in the data.

Context: Even after filtering out flagged sessions (`is_anomaly`, `is_fake`, `is_excluded`), interesting patterns may remain. We also want to validate whether the existing flags are catching the right things.

Technical Notes

- All timestamps are in **UTC**
 - The `session_start` column is a string — parse it as `datetime` before use
 - `session_duration = 0` is valid and means the device was seen only in a single scan cycle (a few seconds). Do not exclude these automatically.
 - The `box_mac` in session data corresponds to `box_macs` (a JSON list) in facility data — you will need to parse the list and match accordingly
 - For Task 3, we will provide the mall floor plan separately once you begin that task
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Questions?

If anything is unclear, please ask before starting. We prefer you spend 30 minutes asking questions upfront rather than building in the wrong direction.